Prometheus Installation.

What is Prometheus?

Prometheus is an **open-source monitoring and alerting tool** designed for:

- Collecting metrics
- Storing time-series data
- Querying metrics
- Triggering alerts

It was originally developed at **SoundCloud** and is now a graduated project under the **Cloud Native Computing Foundation (CNCF)**.

Why Prometheus is Used in DevOps

Modern systems are:

- Distributed
- Containerized
- Microservice-based
- Dynamic (servers spin up/down)

Traditional monitoring tools don't handle this well.

Prometheus is built for:

- Cloud-native environments
- Kubernetes
- Dynamic service discovery

installation of Prometheus using wget . And un tar using the **tar -xvf<file name>**.

```
[root@ip-172-31-82-162 opt]# ls
aws
[root@ip-172-31-82-162 opt]# wget https://github.com/prometheus/prometheus/releases/download/v2.52.0/prometheus-2.52.0.linux-amd64.tar.gz
--2026-02-22 05:38:02--  https://github.com/prometheus/prometheus/releases/download/v2.52.0/prometheus-2.52.0.linux-amd64.tar.gz
Resolving github.com (github.com)... 140.82.112.4
Connecting to github.com (github.com)|140.82.112.4|:443... connected.
HTTP request sent, awaiting response... 302 Found
Location: https://release-assets.githubusercontent.com/github-production-release-asset/6838921/768138e4-e78f-4fb9-b334-959077fdbf25?sp=r&sv=2018-11-09&sr=b&spr=https&s
e=2026-02-22T06%3A28%3A01Z&rscd=attachment%3B+filename%3Dprometheus-2.52.0.linux-amd64.tar.gz&rsct=application%2Foctet-stream&skoid=96c2d410-5711-43a1-aedd-ab1947aa7ab
0&sktid=398a6654-997b-47e9-b12b-9515b896b4de&skt=2026-02-22T05%3A27%3A54Z&ske=2026-02-22T06%3A28%3A01Z&sks=b&skv=2018-11-09&sig=reNN%2F%2Fb5wGMT0%2Brh62zdBKUzHEIRw692k
gTMW9O2%2FuM%3D&jwt=eyJ0eXAiOiJKV1QiLCJhbGciOiJIUzI1NiJ9.eyJpc3MiOiJnaXRodWIuY29tIiwiYXVkIjoicmVsZWFzSlhc3NldHMuZ210aHVidXNlcmNvbnRlbnQuY29tIiwia2V5Ijoia2V5MSIsImV4cC
I6MTc3MTc0MDM4OSwibmJmIjoxNzcxNzM4NTg5LCJwYXRoIjoicmVsZWFzZWFzc2V0cHJvZHVjdGlvbi5ibG9iLmNvcmUud2luZG93cy5uZXQifQ.0xx4ejX3aBBdjXTaa7RzWUk5kQZNwC18ZFxgrK8EySs&response-c
ontent-disposition=attachment%3B%20filename%3Dprometheus-2.52.0.linux-amd64.tar.gz&response-content-type=application%2Foctet-stream [following]
--2026-02-22 05:38:02--  https://release-assets.githubusercontent.com/github-production-release-asset/6838921/768138e4-e78f-4fb9-b334-959077fdbf25?sp=r&sv=2018-11-09&s
r=b&spr=https&se=2026-02-22T06%3A28%3A01Z&rscd=attachment%3B+filename%3Dprometheus-2.52.0.linux-amd64.tar.gz&rsct=application%2Foctet-stream&skoid=96c2d410-5711-43a1-a
edd-ab1947aa7ab0&sktid=398a6654-997b-47e9-b12b-9515b896b4de&skt=2026-02-22T05%3A27%3A54Z&ske=2026-02-22T06%3A28%3A01Z&sks=b&skv=2018-11-09&sig=reNN%2F%2Fb5wGMT0%2Brh62
zdBKUzHEIRw69ZkgTMW9O2%2FuM%3D&jwt=eyJ0eXAiOiJKV1QiLCJhbGciOiJIUzI1NiJ9.eyJpc3MiOiJnaXRodWIuY29tIiwiYXVkIjoicmVsZWFzSlhc3NldHMuZ210aHVidXNlcmNvbnRlbnQuY29tIiwia2V5Ijo
ia2V5MSIsImV4cCI6MTc3MTc0MDM4OSwibmJmIjoxNzcxNzM4NTg5LCJwYXRoIjoicmVsZWFzZWFzc2V0cHJvZHVjdGlvbi5ibG9iLmNvcmUud2luZG93cy5uZXQifQ.0xx4ejX3aBBdjXTaa7RzWUk5kQZNwC18ZFxgrK8
EySs&response-content-disposition=attachment%3B%20filename%3Dprometheus-2.52.0.linux-amd64.tar.gz&response-content-type=application%2Foctet-stream
Resolving release-assets.githubusercontent.com (release-assets.githubusercontent.com)... 185.199.111.133, 185.199.109.133, 185.199.108.133, ...
Connecting to release-assets.githubusercontent.com (release-assets.githubusercontent.com)|185.199.111.133|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 104659766 (100M) [application/octet-stream]
Saving to: 'prometheus-2.52.0.linux-amd64.tar.gz'

prometheus-2.52.0.linux-amd64.tar.gz      100%[===================================================================================>]  99.81M   326MB/s    in 0.3s

2026-02-22 05:38:03 (326 MB/s) - 'prometheus-2.52.0.linux-amd64.tar.gz' saved [104659766/104659766]

[root@ip-172-31-82-162 opt]# ls
aws  prometheus-2.52.0.linux-amd64.tar.gz
[root@ip-172-31-82-162 opt]# tar -xvf
aws/                                     prometheus-2.52.0.linux-amd64.tar.gz
[root@ip-172-31-82-162 opt]# tar -xvf prometheus-2.52.0.linux-amd64.tar.gz
```

i-01f5b5dcc9ec82528 (prometheus)

PublicIPs: 52.87.133.216 PrivateIPs: 172.31.82.162

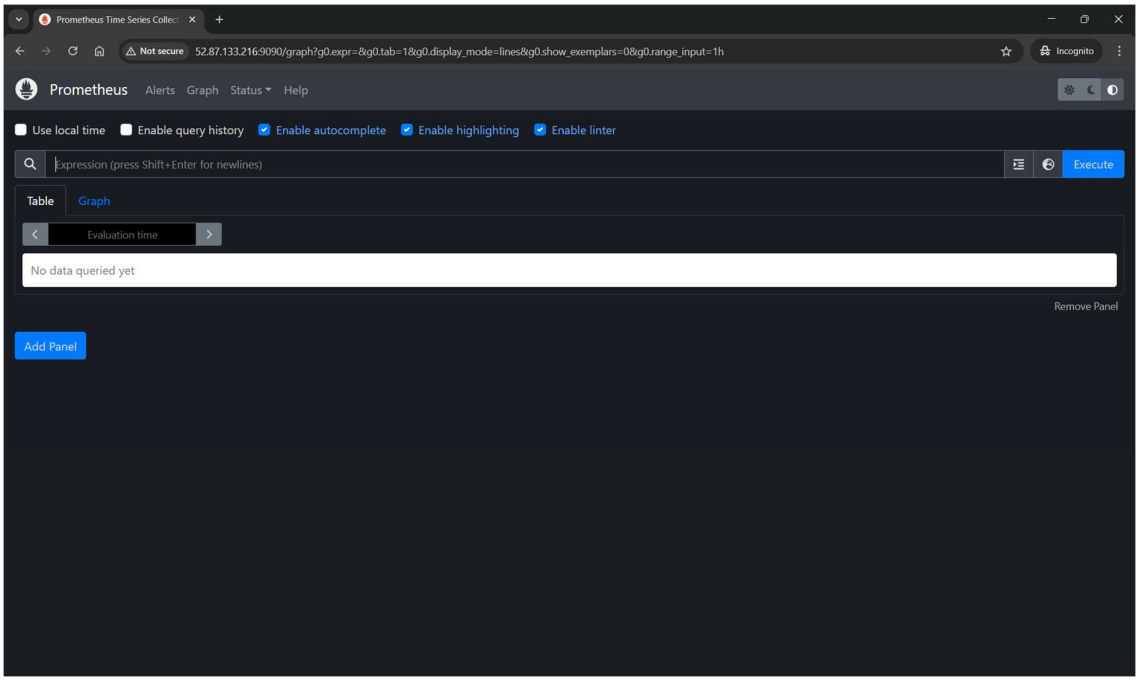
We have to start the prometheus by **./prometheus**.

```
[root@ip-172-31-82-162 prometheus-2.52.0.linux-amd64]# ./prometheus
ts=2026-02-22T05:40:10.868Z caller=main.go:573 level=info msg="No time or size retention was set so using the default time retention" duration=15d
ts=2026-02-22T05:40:10.868Z caller=main.go:617 level=info msg="Starting Prometheus Server" mode=server version="(version=2.52.0, branch=HEAD, revision=879d80922a227c37
df502e7315fad8ceb10a986d)"
ts=2026-02-22T05:40:10.868Z caller=main.go:622 level=info build_context="(go=go1.22.3, platform=linux/amd64, user=root@1b4f4c206e41, date=20240508-21:56:43, tags=netgo
,builtinassets,stringlabels)"
ts=2026-02-22T05:40:10.868Z caller=main.go:623 level=info host_details="(Linux 6.1.161-183.298.amzn2023.x86_64 #1 SMP PREEMPT_DYNAMIC Tue Jan 27 05:01:22 UTC 2026 x86_
64 ip-172-31-82-162.ec2.internal (none))"
ts=2026-02-22T05:40:10.868Z caller=main.go:624 level=info fd_limits="(soft=65535, hard=65535)"
ts=2026-02-22T05:40:10.868Z caller=main.go:625 level=info vm_limits="(soft=unlimited, hard=unlimited)"
ts=2026-02-22T05:40:10.871Z caller=web.go:568 level=info component=web msg="Start listening for connections" address=0.0.0.0:9090
ts=2026-02-22T05:40:10.872Z caller=main.go:1129 level=info msg="Starting TSDB ..."
ts=2026-02-22T05:40:10.873Z caller=tls_config.go:313 level=info component=web msg="Listening on" address=[::]:9090
ts=2026-02-22T05:40:10.873Z caller=tls_config.go:316 level=info component=web msg="TLS is disabled." http2=false address=[::]:9090
ts=2026-02-22T05:40:10.874Z caller=head.go:616 level=info component=tsdb msg="Replaying on-disk memory mappable chunks if any"
ts=2026-02-22T05:40:10.875Z caller=head.go:703 level=info component=tsdb msg="On-disk memory mappable chunks replay completed" duration=1.23µs
ts=2026-02-22T05:40:10.875Z caller=head.go:711 level=info component=tsdb msg="Replaying WAL, this may take a while"
ts=2026-02-22T05:40:10.875Z caller=head.go:783 level=info component=tsdb msg="WAL segment loaded" segment=0 maxSegment=0
ts=2026-02-22T05:40:10.875Z caller=head.go:820 level=info component=tsdb msg="WAL replay completed" checkpoint_replay_duration=22.362µs wal_replay_duration=190.499µs w
bl_replay_duration=196ns chunk_snapshot_load_duration=0s mmap_chunk_replay_duration=1.23µs total_replay_duration=265.84µs
ts=2026-02-22T05:40:10.878Z caller=main.go:1150 level=info fs_type=XFS_SUPER_MAGIC
ts=2026-02-22T05:40:10.878Z caller=main.go:1153 level=info msg="TSDB started"
ts=2026-02-22T05:40:10.878Z caller=main.go:1335 level=info msg="Loading configuration file" filename=prometheus.yml
ts=2026-02-22T05:40:10.881Z caller=main.go:1372 level=info msg="Completed loading of configuration file" filename=prometheus.yml totalDuration=3.537515ms db_storage=1.
32µs remote_storage=1.891µs web_handler=657ns query_engine=996ns scrape=3.168952ms scrape_sd=29.405µs notify=21.445µs notify_sd=5.931µs rules=1.936µs tracing=6.052µs
ts=2026-02-22T05:40:10.881Z caller=main.go:1114 level=info msg="Server is ready to receive web requests."
ts=2026-02-22T05:40:10.881Z caller=manager.go:163 level=info component="rule manager" msg="Starting rule manager..."
```

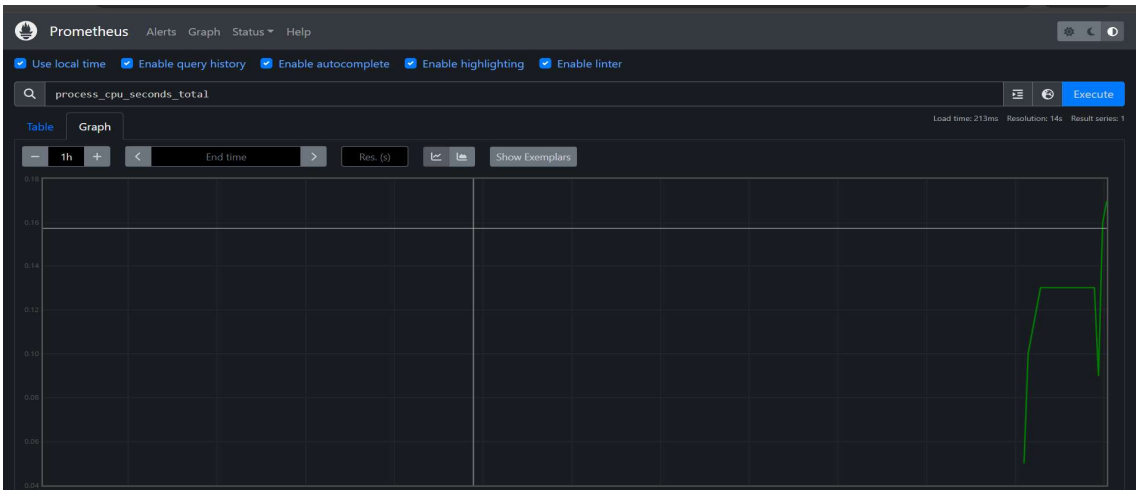
i-01f5b5dcc9ec82528 (prometheus)

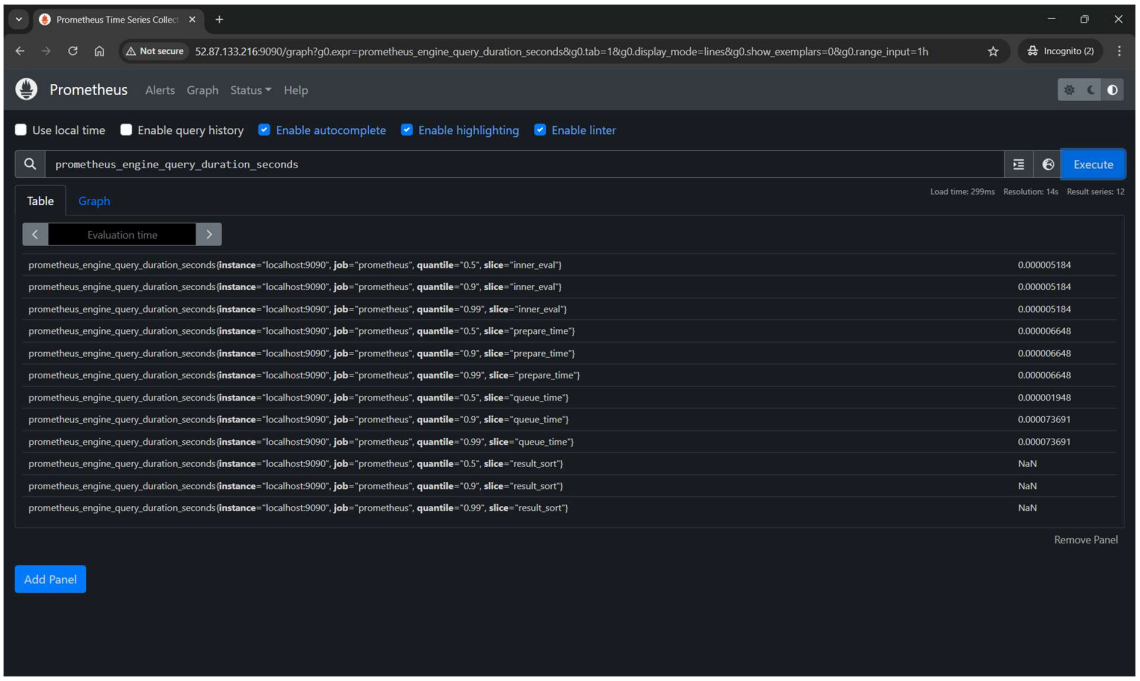
PublicIPs: 52.87.133.216 PrivateIPs: 172.31.82.162

The hosted page of the Prometheus in **<hostpublicip>:9090**.



Process management command has been successfully executed.




Prometheus Time Series Collect
Not secure
52.87.133.216:9090/graph?g0.expr=prometheus_engine_query_duration_seconds&g0.tab=1&g0.display_mode=lines&g0.show_exemplars=0&g0.range_input=1h
Incognito (2)
Prometheus
Alerts
Graph
Status
Help
Use local time
Enable query history
Enable autocomplete
Enable highlighting
Enable linter
prometheus_engine_query_duration_seconds
Execute
Table
Graph
Load time: 299ms
Resolution: 14s
Result series: 12
Evaluation time
prometheus_engine_query_duration_seconds{instance="localhost:9090", job="prometheus", quantile="0.5", slice="inner_eval"} 0.000005184
prometheus_engine_query_duration_seconds{instance="localhost:9090", job="prometheus", quantile="0.9", slice="inner_eval"} 0.000005184
prometheus_engine_query_duration_seconds{instance="localhost:9090", job="prometheus", quantile="0.99", slice="inner_eval"} 0.000005184
prometheus_engine_query_duration_seconds{instance="localhost:9090", job="prometheus", quantile="0.5", slice="prepare_time"} 0.000006648
prometheus_engine_query_duration_seconds{instance="localhost:9090", job="prometheus", quantile="0.9", slice="prepare_time"} 0.000006648
prometheus_engine_query_duration_seconds{instance="localhost:9090", job="prometheus", quantile="0.99", slice="prepare_time"} 0.000006648
prometheus_engine_query_duration_seconds{instance="localhost:9090", job="prometheus", quantile="0.5", slice="queue_time"} 0.000001948
prometheus_engine_query_duration_seconds{instance="localhost:9090", job="prometheus", quantile="0.9", slice="queue_time"} 0.000073691
prometheus_engine_query_duration_seconds{instance="localhost:9090", job="prometheus", quantile="0.99", slice="queue_time"} 0.000073691
prometheus_engine_query_duration_seconds{instance="localhost:9090", job="prometheus", quantile="0.5", slice="result_sort"} NaN
prometheus_engine_query_duration_seconds{instance="localhost:9090", job="prometheus", quantile="0.9", slice="result_sort"} NaN
prometheus_engine_query_duration_seconds{instance="localhost:9090", job="prometheus", quantile="0.99", slice="result_sort"} NaN
Remove Panel
Add Panel